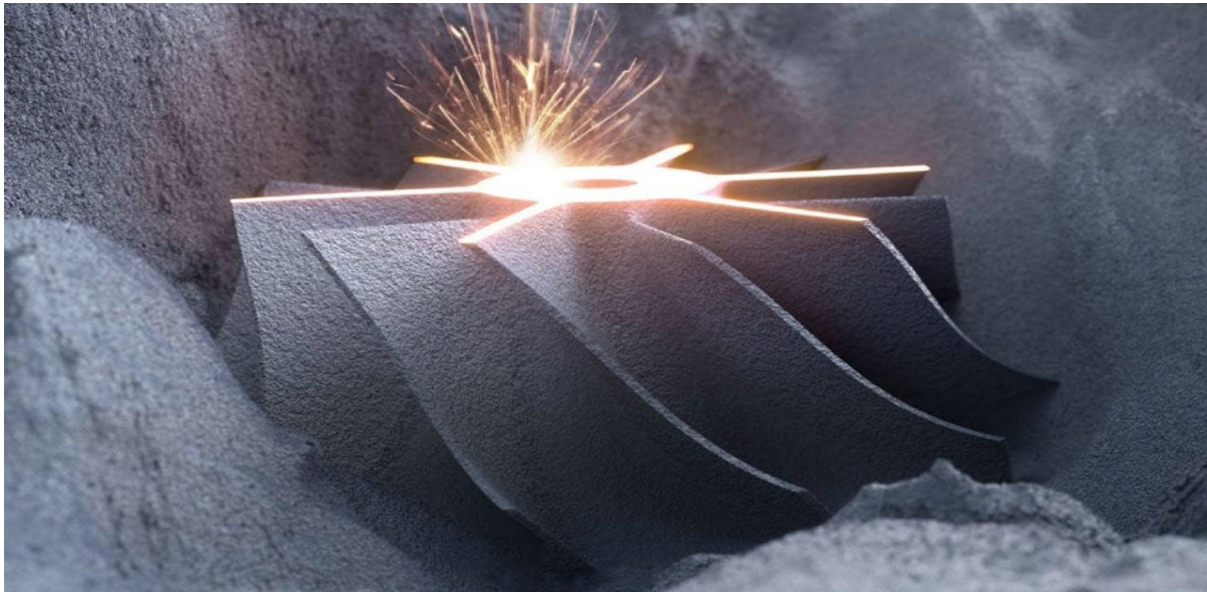




COURSE CONTENT

- 2.1.1 Introduction**
- 2.1.2 Selective Laser Sintering (SLS)**
- 2.1.3 SLS Materials**
- 2.1.4 Comparison of Solid, Liquid and Powder based AM**
- 2.1.5 3D Printing / Binder Jetting**
- 2.1.6 3DP Materials**
- 2.1.7 Laser Engineered Net Shaping (LENS)**
- 2.1.8 LENS Materials**
- 2.1.9 Applications of LENS**

Prof. Shiva Krishna J

Dept. of Mechanical & Automation Engg., Agnel Institute of Technology and Design, Assagao

2.1.1 INTRODUCTION

Powder-based AM processes represent one of the most widely adopted approaches for producing complex and high-performance components directly from digital models. These processes rely on fine powders of metals, polymers, or ceramics as the raw material, which are selectively fused or deposited to build parts layer by layer. Owing to their capability to produce intricate geometries, high material utilization, and suitability for both prototyping and functional end-use applications, powder-based AM processes have gained significant industrial importance.

Broadly, powder-based AM processes are classified into two main categories:

1. Powder Bed Fusion (PBF): In PBF processes, a thin layer of powder is spread across the build platform, and thermal energy sources such as a laser or an electron beam are employed to selectively fuse the powder according to the cross-sectional geometry of the part. Initially developed for fabricating plastic prototypes using laser scanning, PBF has since evolved to include metals and ceramics, offering high precision, fine resolution, and excellent surface finish.

Various AM systems fall under this category are:

- Selective Laser Sintering (SLS) – Fuses polymer powders (e.g.: Nylon).
- Selective Laser Melting (SLM) – Fully melts metal powders (e.g., titanium, stainless steel).
- Direct Metal Laser Sintering (DMLS) – Similar to SLM, often used for alloys.
- Electron Beam Melting (EBM) – Uses electron beams to melt metal powders (commonly titanium).
- Multi Jet Fusion (MJF) – Uses fusing and detailing agents and infrared heating on polymer powders.

2. Binder Jetting: In Binder Jetting processes, a thin layer of powder is spread across the build platform, and a liquid binder is selectively deposited using an inkjet printhead to join powder particles according to the cross-sectional geometry of the part. The process is carried out at room temperature, unlike PBF, and does not require thermal energy to fuse the material. Binder Jetting is particularly suitable for rapid concept models, full-color prototypes, and low-cost production of metal or ceramic parts after post-processing such as sintering or infiltration.

Various AM systems fall under this category:

- Z Corporation's 3DP – Uses plaster or gypsum-based powders with colored binder for full-color concept models.
- Voxeljet – Large-scale binder jetting systems for sand molds and architectural models.

3. Directed Energy Deposition (DED): In contrast, DED processes involve the simultaneous feeding of powder (or) sometimes wire material and the application of a focused thermal energy source - commonly a laser, electron beam, (or) plasma arc to melt the material as it is deposited. The process is highly flexible and is particularly suitable for repairing, adding features to existing components, and producing large, near-net-shape parts.

AM systems under DED are:

- Laser Engineered Net Shaping (LENS) – Delivers metal powder into a Laser beam for melting.
- Electron Beam Additive Manufacturing (EBAM) – Uses an Electron beam to melt metal wire feedstock.
- Wire Arc Additive Manufacturing (WAAM) – Uses an Electric arc to melt wire material.
- Direct Metal Deposition (DMD) – Similar to LENS, Laser-based with powder feeding.

In the present chapter, we will focus on one representatives of these categories: SLS under PBF, 3DP under Binder Jetting and LENS under DED. These processes will be discussed in detail with respect to their working principles, materials, advantages, limitations, and applications.

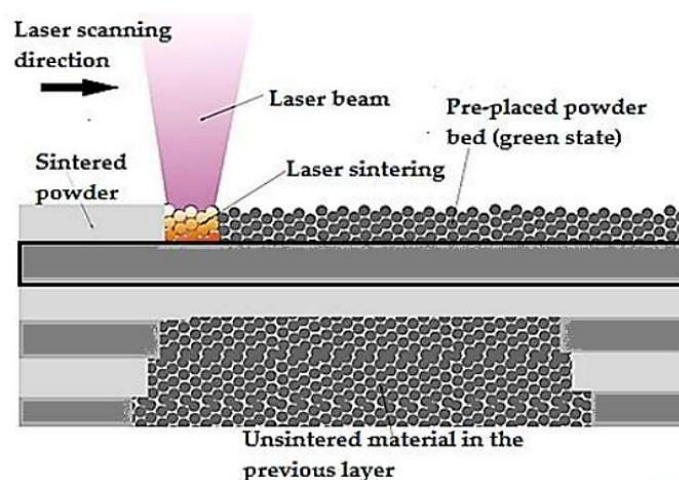
2.1.2 SELECTIVE LASER SINTERING (SLS)

SLS was developed at the University of Texas at Austin and commercialized by DTM Corporation in 1987 with support from B.F. Goodrich. DTM held the worldwide exclusive license to market SLS technology until the company was acquired by 3D Systems in August 2001.

In SLS, a part is built layer by layer by selectively sintering powdered material with a high-powered CO₂ laser. A thin layer of powder is spread onto the build platform, and the laser scans the cross-sectional geometry of the CAD model. The laser energy raises the powder temperature above its glass-transition point, softening particles enough for them to bond together and to the previous layer. This process is repeated until the entire part is completed.

SINTERING

Sintering involves heating powder particles above their glass-transition temperature, where the material softens but does not fully melt. In this semi-solid state, particles deform at their contact points, allowing them to bond and form a solid structure without entering a fully molten phase. This reduces the risk of distortion or warping that can occur with complete melting. After cooling, the bonded powder forms a stable matrix with a density close to that of the raw material.

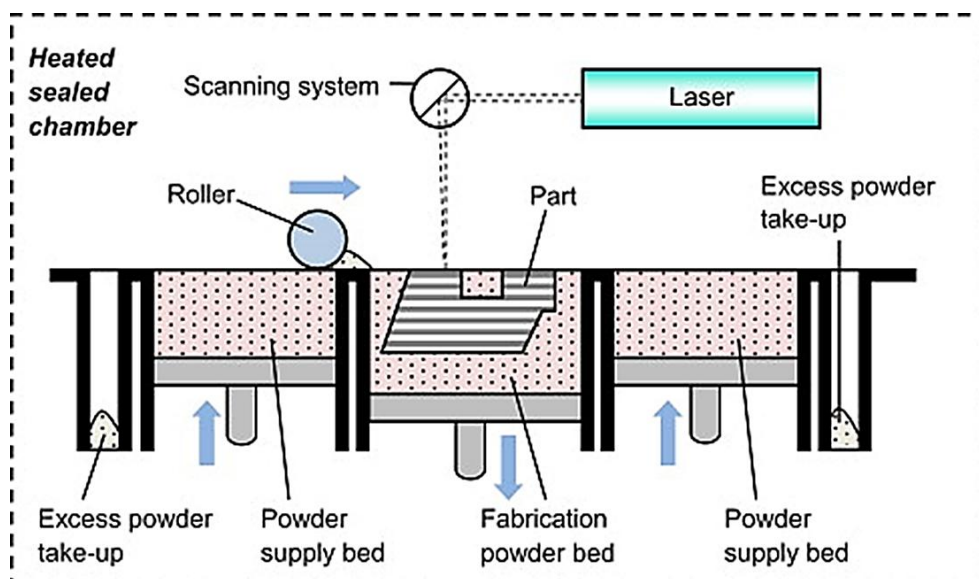


Principle of Sintering

Because the process requires heating the particles to near their transition temperature, SLS consumes significantly more energy approximately 300 to 500 times higher than photopolymerization processes. To reduce laser power requirements, auxiliary heaters preheat the powder bed close to the sintering temperature. An inert gas environment (such as nitrogen) is essential to prevent oxidation or explosion of fine powders, and chamber cooling systems are used to stabilize the build environment.

HARDWARE

- Fabrication Powder Bed: Also referred to as Build Platform where the part is built layer by layer. It moves in – Z direction after printing each layer.
- Powder Supply Beds: Two reservoirs on either side of the build platform that move in the +Z direction to feed fresh powder.
- Excess Powder Take-up Chamber: Collects surplus powder displaced during recoating.
- Roller: Spreads a thin, uniform powder layer from the supply bed onto the build platform.
- CO₂ Laser: Provides the energy required to sinter the powder particles.
- Optical Scanning System: Directs and controls the laser beam path with precision, guided by the CAD model.



Hardware and Principle of SLS

PROCESS

A 3D CAD model in STL file format after slicing is sent to the machine for building the part layer by layer. A thin layer of heat-fusible powder is deposited onto the part building chamber. The bottom-most cross-sectional slice of the CAD part under fabrication is selectively drawn on the layer of powder by a heat-generating CO₂ laser. The interaction of the laser beam with the powder elevates the temperature to the point of melting, fusing the powder particles to form a solid mass. When the cross-section is completely drawn, an additional layer of powder is deposited via a roller mechanism on top of the previously

scanned layer. The process is repeated, with each layer fusing to the layer below it and until the part is completed.

The unfused powder during processing serves as a built-in support structure. There is no need to create support structures and thus no support structure to remove when the part is completed.

After the part is built, the powder cake is taken to the Breakout station table, where the excess powder is removed from the parts manually with air brushes and vacuum. The parts may then require some post-processing (or) secondary finishing, such as sanding, lacquering and painting, depending upon the application of the prototype built.

ADVANTAGES:

1. **Good part stability:** Parts are created within a precise controlled environment. The process and materials provide for directly produced functional parts to be built.
2. **Wide range of processing materials:** A wide range of materials including Nylon, TPU, TPE thus providing flexibility and a wide scope of functional applications.
3. **No part supports required:** The system does not require CAD developed support structures.
4. **Little post-processing required:** The finishing of the part is reasonably fine and requires only minimal post-processing such as particle blasting and sanding.
5. **No post-curing required:** The completed laser sintered part is generally solid enough and does not require further curing.
6. **Batch production capability:** Multiple parts can be produced simultaneously.

LIMITATIONS:

1. **Large physical size of the unit:** The system requires a relatively large space to house it. Apart from this, additional storage space is required to house the inert gas tanks used for each build.
2. **High power consumption:** The system requires high power consumption due to the high wattage of the laser required to sinter the powder particles together.
3. **Surface finish limitations:** As-built parts often have a rougher texture due to powder granularity.

2.1.3 SLS MATERIALS

In SLS, a wide range of materials can be used depending on the intended application, ranging from polymers for functional prototypes to metals and ceramics for high-performance and specialized parts. Polymers, particularly Nylon and its reinforced composites, are the most commonly used due to their balance of strength, toughness, and processability. Metals are processed using SLS variants such as DMLS and SLM to produce dense, functional, and load-bearing components, while ceramics are primarily explored for high-temperature, wear-resistant, and biomedical applications. The following table categorizes these materials along with their key properties and typical applications.

SLS Materials – Composition, Properties, and Applications

Category	Material	Key Properties	Applications
Polymers	Nylon (PA11, PA12)	High strength, toughness, wear & chemical	Automotive components, housings, tooling, medical

		resistance	models
	Glass Filled Nylon	High stiffness, dimensional stability, wear resistance	Structural parts, enclosures, jigs & fixtures
	Aluminum Filled Nylon	Metallic look, high stiffness, thermal stability	Automotive prototypes, functional & heat-resistant parts
	Carbon Fiber Reinforced Nylon	High stiffness-to-weight ratio, excellent strength	Aerospace, motorsport, lightweight tooling
	PEEK / PEKK	High strength, thermal & chemical resistance, biocompatible	Aerospace, high-performance parts, medical implants
	Thermoplastic Polyurethane (TPU)	Flexibility, elasticity, impact resistance	Footwear, seals, gaskets, flexible prototypes
Metals	Stainless Steel	High strength, corrosion resistance	Medical tools, aerospace parts, functional prototypes
	Aluminum	Lightweight, strength-to-weight ratio, thermal conductivity	Automotive, aerospace, heat exchangers
	Titanium	High strength, lightweight, biocompatibility	Medical implants, aerospace structures
	Cobalt Chrome	Wear resistance, biocompatibility, high-temperature stability	Dental implants, turbine blades, orthopedic parts
	Nickel Alloys	High-temperature strength, oxidation & corrosion resistance	Jet engines, gas turbines, heat exchangers
Ceramics	Silica (SiO ₂)	High-temp resistance, insulating	Investment casting molds, heat shields
	Alumina (Al ₂ O ₃)	Hardness, wear resistance, electrical insulation	Cutting tools, medical implants, substrates
	Zirconia (ZrO ₂)	High toughness, thermal shock resistance	Dental crowns, wear parts, cutting inserts
	Hydroxyapatite (Ca ₁₀ (PO ₄) ₆ (OH) ₂)	Biocompatible, bone-like	Bone scaffolds, biomedical implants

2.1.4 COMPARISON OF SOLID, LIQUID, AND POWDER BASED AM

AM encompasses a variety of technologies that differ in the state of the material as solid, liquid, or powder. Solid-based processes, such as FDM, extrude thermoplastic filaments to build parts layer by layer. Liquid-based processes, like SLA, cure photopolymer resins using light to create high-resolution models. Powder-based processes, such as SLS, fuse powdered materials using thermal energy to produce functional prototypes and end-use parts. Each method has distinct material requirements, process characteristics, and application areas, making it important to compare them in terms of material type, mechanical properties, and typical uses.

The following table shows the Comparison of Solid, Liquid, and Powder Based AM Technologies: FDM, SLA, and SLS

Comparison of FDM, SLA, and SLS AM Technologies

Criteria	FDM	SLA	SLS
Material	Thermoplastic filaments: PLA, ABS, PETG, Nylon, PC etc.	Photopolymer resins (acrylic or epoxy based)	Powdered polymers Nylon, TPU powders and ceramics.
Layer Thickness (Z-axis resolution)	0.1 – 0.3 mm (100–300 μm)	0.025 – 0.1 mm (25–100 μm)	0.06 – 0.15 mm (60–150 μm)
Surface Finish	Noticeable layer lines; stepped effect especially on curved surfaces	Very high detail; smooth surfaces; sharp features	Surface is grainy/matte because of powder particle sintering;
Support Structures	Required for overhangs, support removal and surface finishing needed.	Required; supports are usually fine, support removal and post curing.	Generally not required: unsintered powder around the part acts as support.
Strength	Good, but parts are weaker in layer bonding direction.	Resins tend to be more brittle; strength lower than compared to FDM (or) SLS	Generally stronger. Better impact resistance, thermal performance, durability.
Dimensional Accuracy / Precision	Less precision: typically, tolerance is $\pm 0.2 - 0.5$ mm	Higher precision: SLA can achieve tighter tolerances ± 0.05 mm	Moderate precision: typical tolerances around $\pm 0.1 - 0.3$ mm,
Build Speed	Moderate; Complex geometries or large parts take time	SLA is slow; DLP (or) MSLA variants can cure whole layers simultaneously, but cleaning, post curing add extra time.	SLS can be efficient for batch production. Also no supports so less wasted time.
Cost (Machine + Materials)	Lowest among the three	Medium to high	Highest

2.1.5 BINDER JETTING / 3D PRINTING (3DP)

The technology is often referred to as 3DP technology and is copyrighted under this name, in which a binder is printed onto a powder bed to form part cross sections.

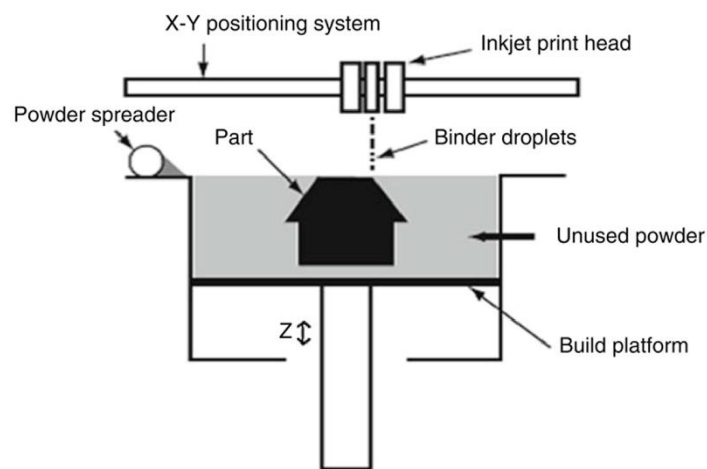
Binder Jetting is an additive manufacturing process in which an industrial printhead selectively deposits a liquid binding agent onto a thin layer of powder particles such as foundry sand, polymer and ceramics to build parts and tooling.

3DP process was developed at the Massachusetts Institute of Technology (MIT) in the early 1990s and was licensed to Z-Corporation for commercialization in 1995. The company became well-known for producing rapid, low-cost concept models, particularly due to its unique capability of printing full-color prototypes. Z Corporation was later acquired by 3D Systems in 2012.

In 3DP, a part is built layer by layer by selectively depositing a liquid binder onto a thin layer of powdered material. An inkjet printhead passes over the powder bed, jetting droplets of binder to glue the particles together in the cross-sectional geometry of the CAD model. Colored inks (CMYK) can be added to the binder to produce vivid, multi-colored models. The process is repeated until the entire part is completed.

BINDING

Since the process relies on inkjet binder deposition, energy requirements are significantly lower compared to processes such as SLS. No lasers, heaters, or inert gas chambers are required. The powder bed itself provides natural support for the part during printing, eliminating the need for additional support structures. This makes the process faster and more economical, but the produced models are generally weaker and more porous compared to laser-sintered parts.



Principle of Binding

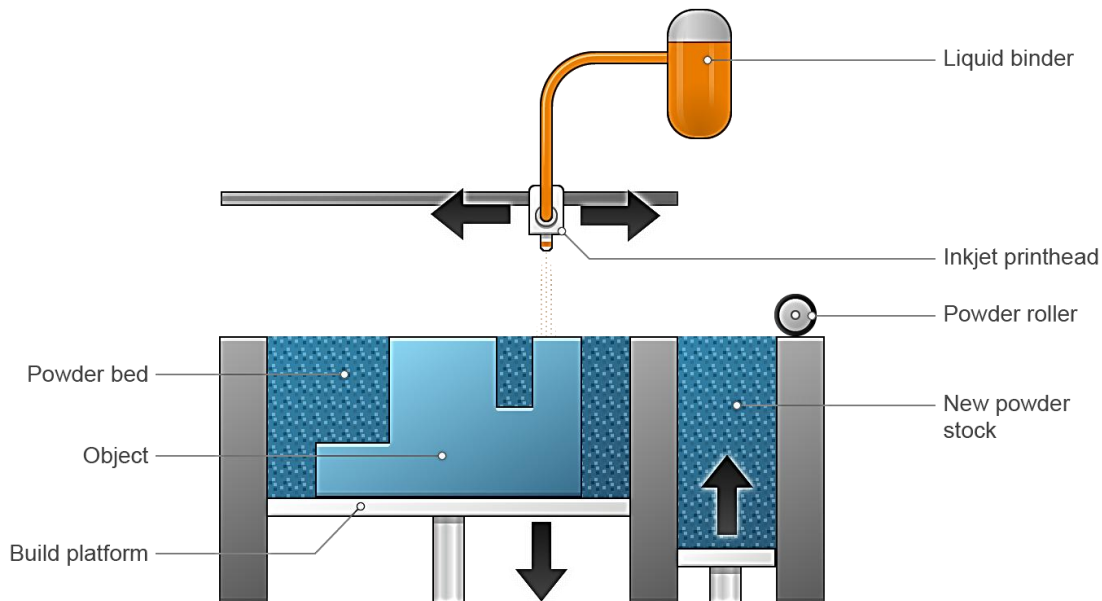
HARDWARE

- **Build platform**: Also referred as Powder Bed where the part is constructed layer by layer on this platform, which moves in $-Z$ direction after each layer is printed.
- **Powder Feed bed**: A supply reservoir that holds the raw powder and raises in $+Z$ direction to supply new material.
- **Powder roller**: Spreads a thin, uniform layer of powder from the feed bed onto the build platform.
- **Inkjet Printhead**: Jets binder droplets and colored inks onto the powder surface to define the part geometry.

PROCESS

A CAD model is first converted into an STL file and sliced into thin layers. The printing process begins with a thin layer of powder being uniformly spread across the build platform. An inkjet printhead then selectively deposits binder onto the powder bed, corresponding to the cross-sectional slice of the CAD model. If color printing is enabled, CMYK inks are simultaneously jetted to produce full-color prototypes. The deposited binder causes the powder particles to adhere together in the defined regions, forming a solidified layer. After each pass, the build platform lowers by one layer thickness while the feed bed

risers to supply fresh powder, which is evenly spread across the surface by a roller. This sequence is repeated layer by layer until the entire part is completed within the surrounding powder bed.



2

Hardware and Principle of Binder Jetting

Once complete, the part remains surrounded by loose, unbound powder, which acts as a natural support. The surrounding powder is removed in a depowdering station using air brushes or vacuum. Post-processing such as infiltration, painting, or sanding may be applied depending on the intended application.

ADVANTAGES

1. ***Low-cost and fast:*** Compared to other AM methods, parts can be printed quickly at relatively low cost.
2. ***Full-color capability:*** Unique ability to print parts in full color using CMYK binder inks.
3. ***No support structures:*** Loose powder provides built-in support, simplifying the process.
4. ***Room-temperature process:*** No lasers, heaters, or inert atmospheres required.
5. ***Educational and visualization use:*** Excellent for concept models, design reviews, and demonstration purposes.

LIMITATIONS

1. ***Low mechanical strength:*** Parts are fragile and unsuitable for functional testing without infiltration.
2. ***Surface porosity:*** Parts are porous and require post-processing for durability or smooth finishes.
3. ***Limited material options:*** Primarily gypsum-based powders, with restricted mechanical performance.

4. **Post-processing dependency:** To achieve durability, infiltration with resins or wax is usually required.
5. **Not suitable for end-use parts:** The process is intended for concept modelling and visualization, not production.

2.1.6 3DP MATERIALS

1. BUILD MATERIAL (In Powder Form):

- Plaster/Gypsum-based powders: The most common build material used by Z Corp printers. These powders are inexpensive, easy to spread, and suitable for absorbing binder.
- Starch-based powders: In earlier systems, starch powders were also used due to their compatibility with water-based binders.
- Composite powders: Some proprietary blends were developed to improve green strength and surface finish.

2. BINDER:

- Typically, a water-based liquid binder jetted through an inkjet printhead.
- Contains additives such as polymers, sugars, or salts that promote adhesion between powder particles.

3. COLOURING AGENTS:

- CMYK inks can be added to the binder, enabling full-color 3D printing.

4. POST-PROCESSING / INFILTRANTS:

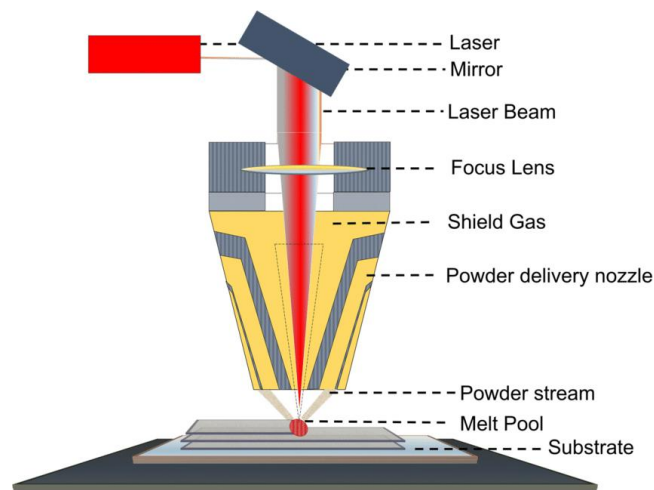
- Cyanoacrylate (super glue): Commonly used to infiltrate parts and give them strength.
- Epoxy resins: Provide higher durability and smoother finishes.
- Wax or paraffin: Used for sealing porous parts.

2.1.7 LASER ENGINEERED NET SHAPING (LENS)

Laser Engineered Net Shaping (LENS) is a process in which near-net-shape metal structures are built of powdered metal layer by layer from CAD design. LENS is a promising metal additive manufacturing process. It prints metal parts using a variety of materials: Cobalt, Aluminum, Stainless steel, Copper Titanium, Nickel and Tool steel using a high-powered Nd:YAG laser. The LENS process is originally developed at Sandia National Laboratories, LENS has been refined and commercialized by Optomec, a privately-held US based company headquartered in New Mexico.

PRINCIPLE

- In LENS, metal powder is fed through nozzles into a molten pool created by a high-power laser on the substrate or previously deposited layer.
- The laser melts the powder as it is deposited, allowing it to solidify immediately and form a dense, near-net-shape part.
- Argon gas is used to deliver the powder and protect the melt pool from contamination.



Principle of LENS

HARDWARE:



Hardware of LENS

1. Laser Unit and Laser Head: LENS machine uses Nd: YAG (Neodymium: Yttrium Aluminium Garnet) laser of 500W – 3kW. The laser head is surrounded by four equi-spaced nozzles through which the powder is deposited exactly at the point where the laser beam is focused. The laser head move in Z-axis up and down.
2. Build Chamber: This chamber is hermetically sealed and is connected to the glove-box with a small door. The system provides two rubber gloves for handling powder/ components and making required arrangements inside the chamber.
3. Build Platform: A platform, rectangular in shape, is provided in the build chamber for horizontal X-Y directional movement. The platform is slotted with threaded holes to attach the substrate of various sizes.

4. Build Substrate: The plate that is used in-between deposition head and build platform is called the substrate. A 2mm thick plate of appropriate material is adequate to be used as substrate.
5. Argon Gas Circulator: Most of the metals tend to oxidize at higher temperatures. To mitigate the issue of oxidation, argon gas needs to be circulated throughout the material deposition process.
6. Powder Hoppers: The LENS machine has two hoppers with a feeder mechanism to supply the powder to the nozzles from the hoppers. The glass doors are used to monitor the quantity of powder in the hoppers.

PROCESS:

- In LENS system the metal powders are delivered and distributed around the circumference of the head either by gravity or by using an inert, pressurized carrier gas, which also shields the melt pool from atmospheric oxygen.
- The metal powder in a slurry generated by the action of a high power beam laser is deposited on the upper surface of a metal substrate previously deposited on a platform.
- The build platform surface moves in XY direction to produce each layer of the object individually. The head moves vertically up each time a single layer is made.
- The part is built layer by layer under the control of software. When complete, the part is removed and can be heat-treated, Hot-Isostatic Pressed, machined, or finished.

ADVANTAGES

1. **High precision and complex geometries**: Can produce near-net-shape metal parts with intricate designs.
2. **Material flexibility**: Supports a wide range of metals and alloys.
3. **Repair and additive capability**: Can repair worn or damaged components by adding material only where needed.
4. **Minimal material wastage**: Material is deposited only where required, reducing scrap.
5. **Rapid prototyping and small-batch production**: Suitable for fast manufacturing of functional parts with good mechanical properties.

DISADVANTAGES

1. **High equipment cost**: LENS machines are expensive to purchase and maintain.
2. **Surface finish issues**: Parts may have rough surfaces and often require post-processing.
3. **Lower deposition rate for large parts**: Slower compared to wire-fed DED for bulk manufacturing.
4. **Skilled operation required**: Precise control of parameters is necessary to avoid defects.
5. **Risk of porosity and residual stresses**: Improper process control can lead to internal defects in parts.

2.1.8 LENS MATERIALS

1. Titanium and Titanium Alloys: Lightweight, high strength, corrosion-resistant; widely used in aerospace and biomedical implants.

2. Stainless Steel: Strong, corrosion-resistant; used for structural components, tooling, and industrial parts.
3. Nickel-based Superalloys: High-temperature strength and oxidation resistance; ideal for turbine blades, aerospace, and energy applications.
4. Cobalt-Chrome Alloys: High wear resistance and biocompatibility; used in medical implants and dental applications.
5. Aluminum Alloys: Lightweight with good strength-to-weight ratio; used in aerospace and automotive components.

2.1.9 APPLICATIONS OF LENS

1. **Aerospace Components**: Manufacturing and repair of turbine blades, engine parts, and structural components.
2. **Biomedical Implants**: Customized bone implants, prosthetics, and dental components.
3. **Tooling and Molds**: Production of complex molds, dies, and repair of worn-out tools.
4. **Repair and Maintenance**: Restoration of high-value or worn-out metal parts in industries like aerospace, automotive, and energy.
5. **Rapid Prototyping**: Creation of functional prototypes and small-batch production of metal parts.